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# Assessment

1.)

| | Mean | S.D | Size |
|---|---|---|---|
| Girls | 89 | 4 | 50 |
| Boys | 82 | 9 | 120 |

Level of significance = 5%
$\alpha = 0.05$

$H_0$ = No significance difference
$H_1$ = Significance difference

$$Z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

$$Z = \frac{89 - 82}{\sqrt{\frac{4^2}{50} + \frac{9^2}{120}}} \Rightarrow \frac{7}{\sqrt{0.32 + 0.675}}$$

$$Z = \frac{7}{0.997} \Rightarrow Z = 7.02$$

Critical value :- 5% significance level
critical value = 1.96

calculate $Z = 7.02$
critical $Z = 1.96$

$7.02 > 1.96$

Reject $H_0$

→ There is significance difference between boys and girls intellegence.

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2.)

| Category | Diagnosed cancer | without cancer | Total |
|---|---|---|---|
| Smokers | 220 | 230 | 550 |
| Non-smokers | 350 | 640 | 990 |
| Total | 680 | 910 | 1590 |

$$E = \frac{\text{Row Total} \times \text{Clomn Total}}{\text{Grand Total}}$$

Smoker & cancer $= \frac{550 \times 680}{1590} = 235.22$

Smoker & No cancer $= \frac{550 \times 910}{1590} = 314.78$

Non-smokers & cancer $= \frac{990 \times 680}{1590} = 423.77$

Non-smokers & No cancer $= \frac{990 \times 910}{1590} = 566.23$

$$\text{Chi-Square} = x^2 = \sum \frac{(O - E)^2}{E}$$

Smoker & cancer $= \frac{(220 - 235.22)^2}{235.22} = 0.98$

Smoker & No cancer $= \frac{(230 - 314.78)^2}{314.78} = 22.83$

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No smoker & cancer $= \frac{(350 - 423.77)^2}{423.77} = 12.83$

No smoker & no cancer $= \frac{(640 - 566.23)^2}{566.23} = 9.61$

Chi-square value :

$\chi^2 = 0.98 + 22.83 + 12.83 + 9.61$
$= 46.25$

5% significance level
critical value $= 3.84$

$\therefore 46.25 > 3.84$
$\therefore$ reject $H_0$

There is a significant relationship between and cancer.